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Predicting myopia in medical students

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What we will cover

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01 · PROBLEM STATEMENT

Four targets, 24 test students

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01

Myopia

Predict myopia:
yes or no

02

Myopia magna

Predict myopia magna:
over 6 dioptres

03

Combo

Predict one of three:
control, myopia or myopia magna

04

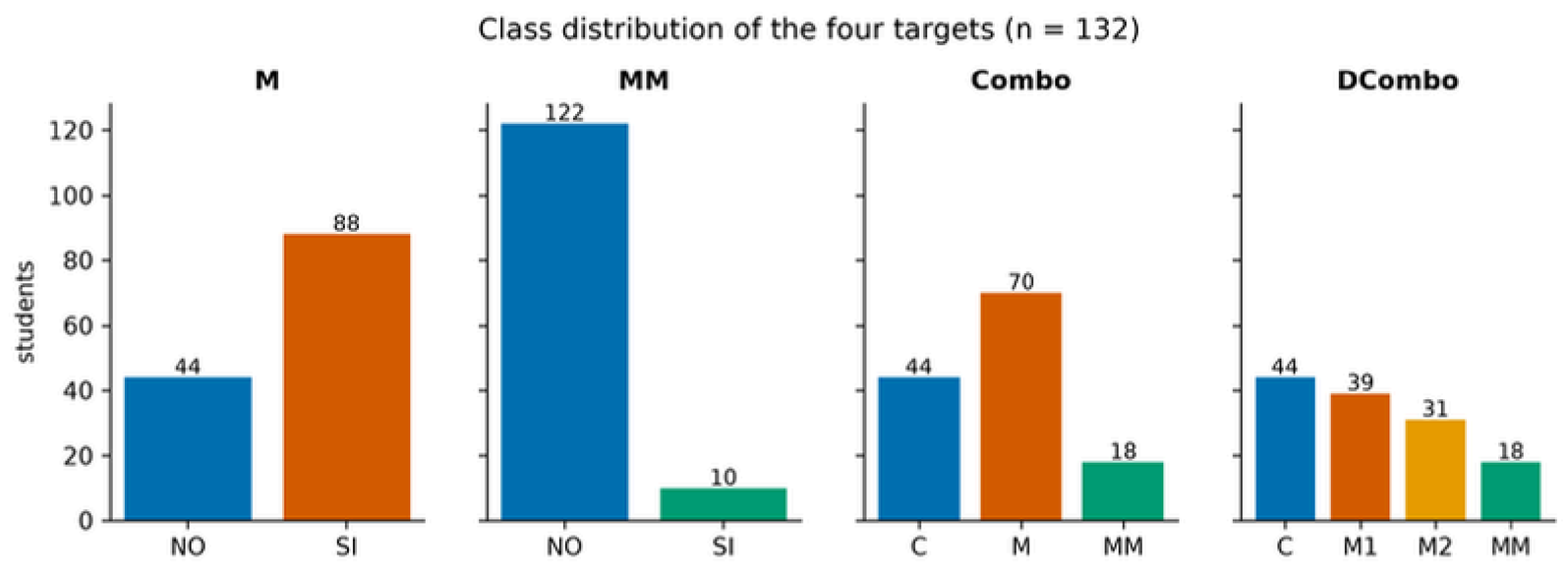
DCombo

Predict one of four:
control, M1, M2 or myopia magna

Why it matters

Myopia magna can lead to retinal disease,
yet it is the rarest class.

Data overview



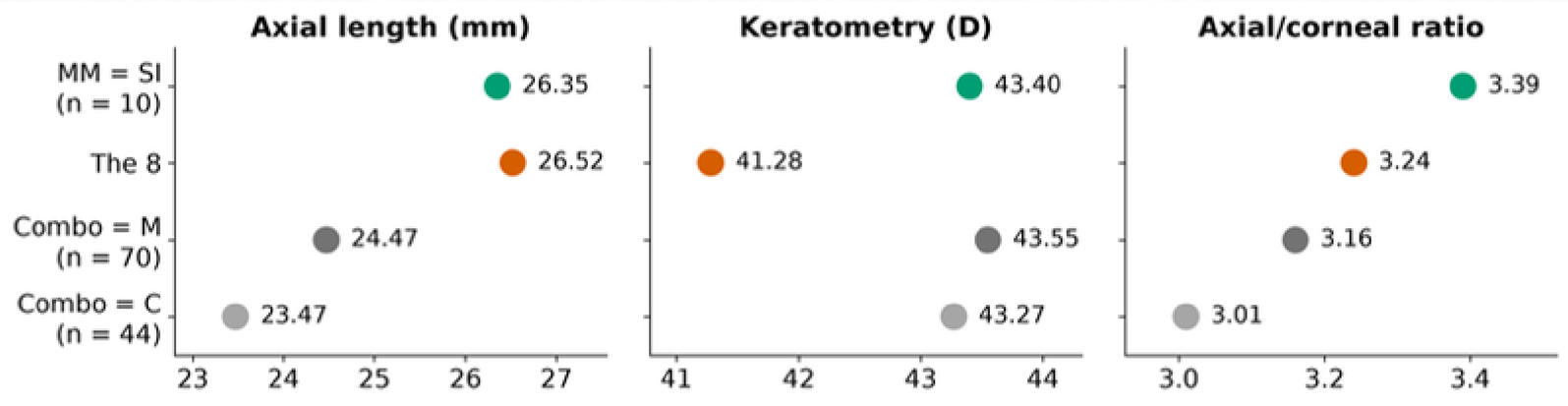
132 training students
49 raw columns

10 myopia magna cases

92% accuracy by always answering no

Axial length is the one clear signal

Inconsistent labels



8 students with Combo = MM, MM = NO

18 magna cases after correction

3/4 targets already say myopia magna

Long eyes, flatter corneas

Cleaning and new features

01

Encoding
49 raw columns become 32 numeric features

02

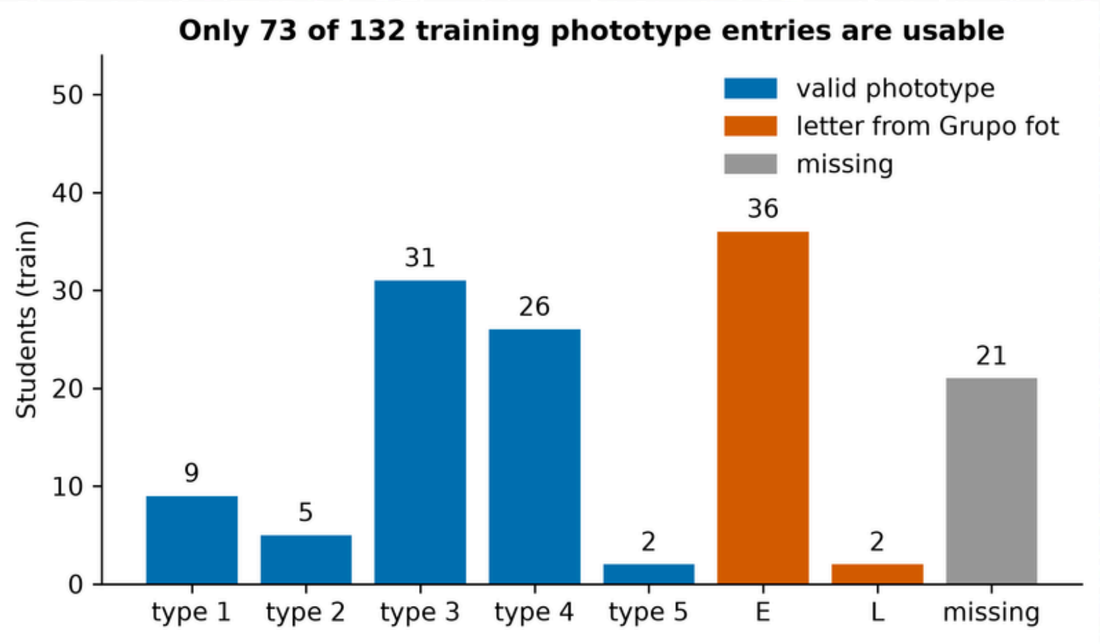
Dropped
phototype: 45% unusable, mixed up with Grupo fot

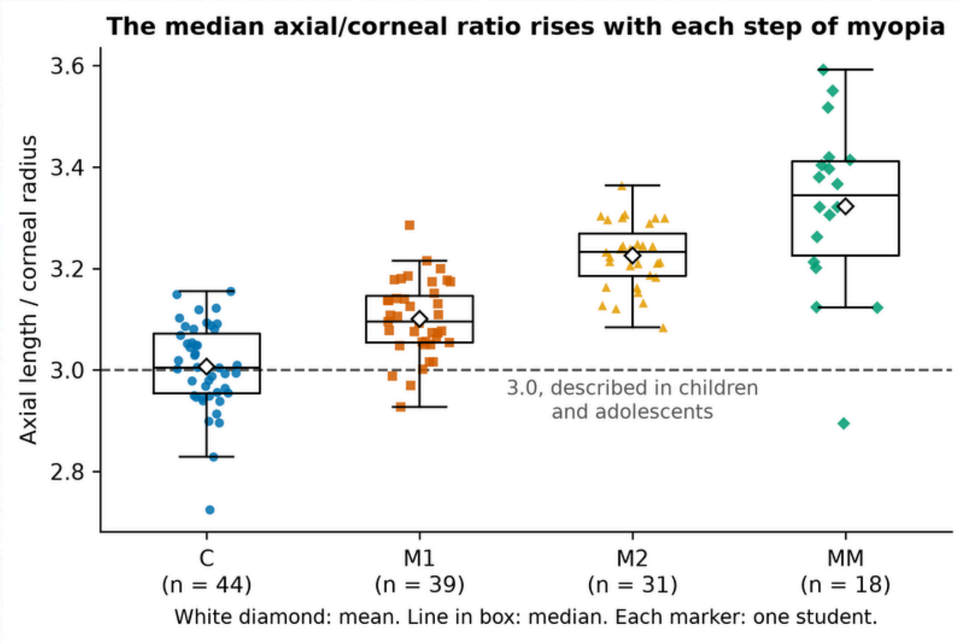
03

Scaling
imputer and scaler fitted inside each training fold

04

New feature
axial length / corneal radius





Class imbalance

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01

Stratified folds

every fold keeps every class, even the rare magna class

02

Balanced weights

rare classes weigh as much in training

03

Right metrics

balanced accuracy and macro-F1, accuracy reported

04

Threshold tested

a tuned MM cut-off did not beat 0.5

The trap

Always answering no looks accurate, but finds no magna case.

Validation

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01 **Pipeline**
Imputer and scaler refitted on each training fold: no leakage

02 **Repeated stratified CV**
5 folds × 10 repeats; we report the spread, not one number

03 **Nested CV**
An inner 3-fold grid search picks the hyperparameters

04 **Test set used once**
Only to predict the final labels; nothing tuned on it

Candidate models

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01

Logistic regression

on axial length alone,
then on all 32 features

02

SVM

RBF kernel for
curved boundaries

03

Random forest

many independent trees,
averaged

04

XGBoost

shallow trees built in sequence;
not seen in class

Two more ideas

The axial/corneal ratio from the literature, and an ordinal model for DCombo.

Decision rule

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01

Baseline first

The better logistic baseline is the bar to beat

02

Rule written first

Clear gain: mean paired difference above one SD

03

Check again

The winner must pass again on new folds

Route A Better balanced accuracy, macro-F1 not worse

Route B Same balanced accuracy, better macro-F1 and accuracy

How we used generative AI

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01

Code

Claude Code wrote much of the code and notebook text

02

Ideas

Conceptual answers and proposed approaches

03

Reviews

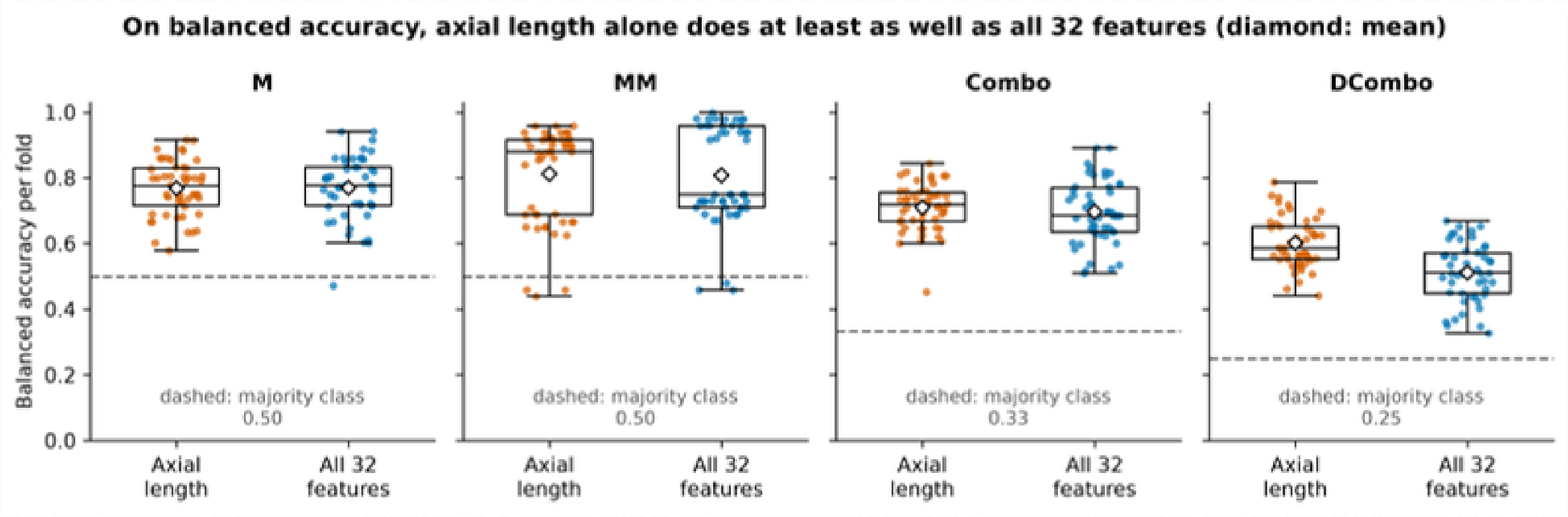
Audits of methodology and charts

04

Decisions

Taken by the team

Baselines



01

Majority class:
no magna found

02

Axial length alone:
a strong baseline

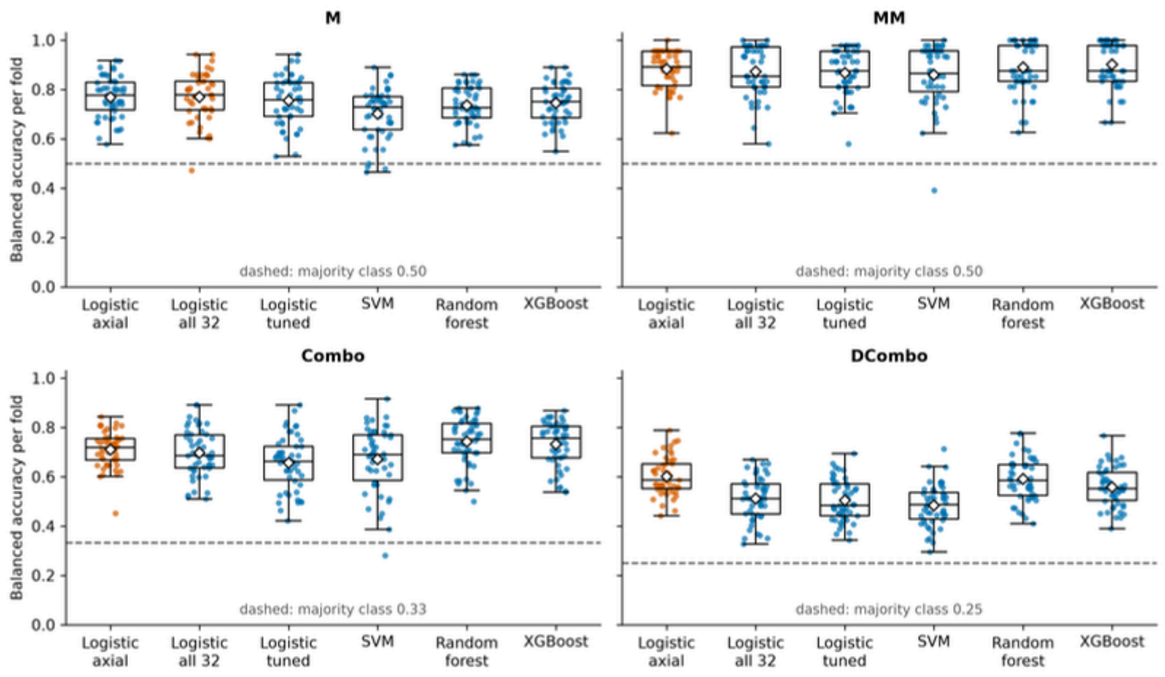
03

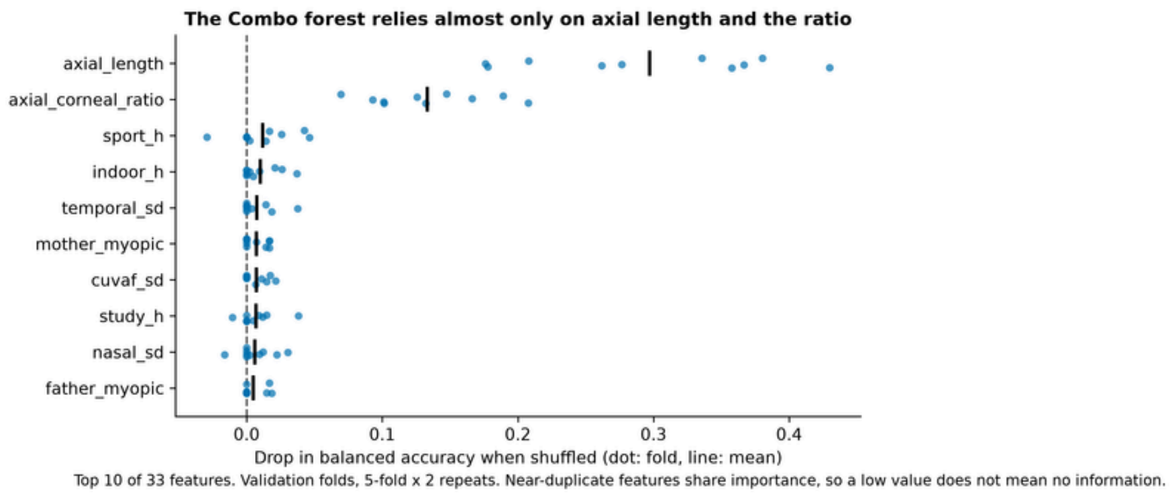
All 32 features:
no better on
balanced accuracy

04

The better one
is the bar to beat

Model comparison





Tree ensembles won on MM and Combo; the **ratio helped M, Combo and DCombo.**

Right: the Combo forest relies on axial length and the ratio.

Final models

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01

M

Logistic regression on axial length + ratio

02

MM

XGBoost on the 32 features, threshold 0.5

03

Combo

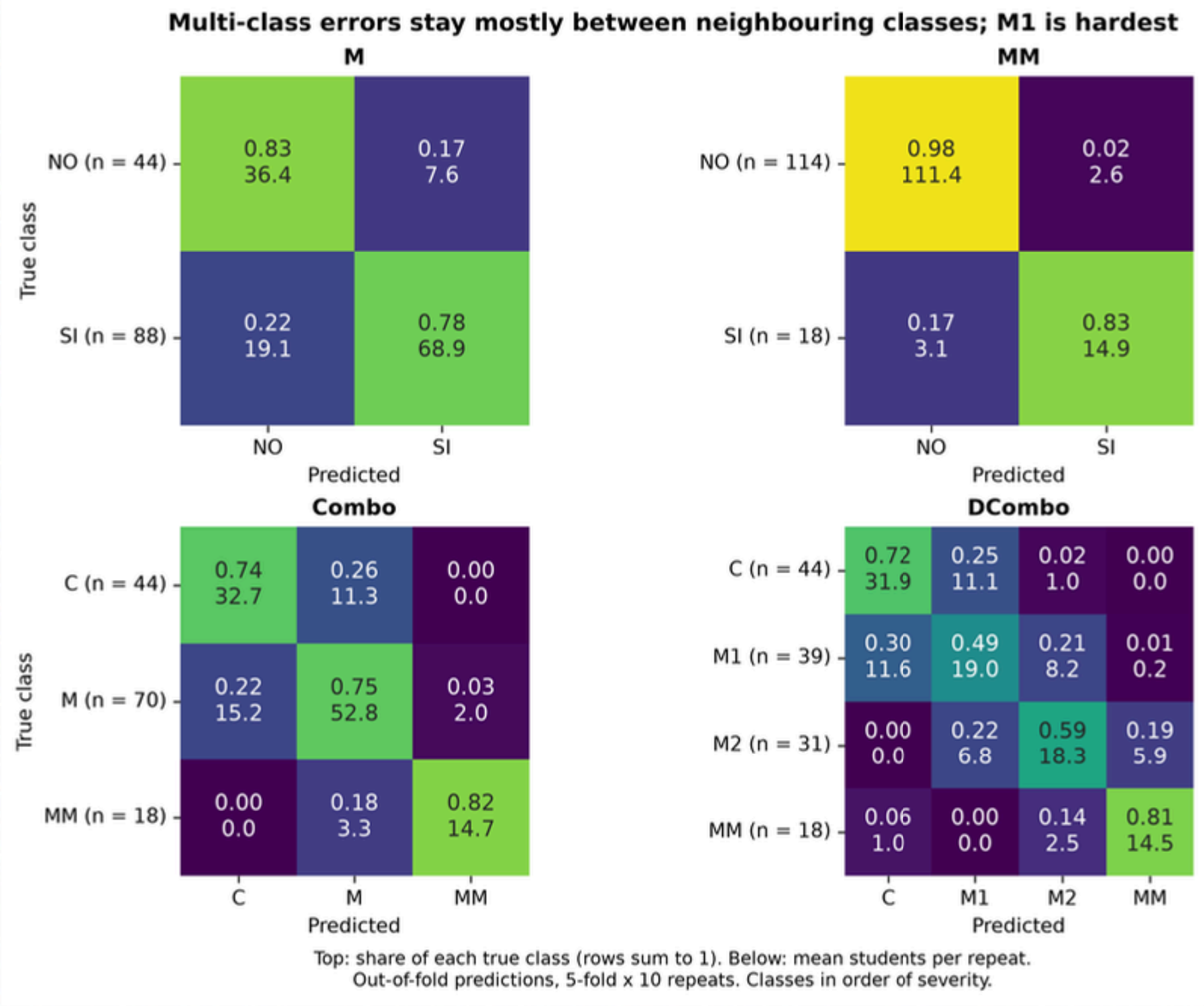
Random forest on the 32 features + ratio

04

DCombo

Logistic regression on axial length + ratio

Where the models fail



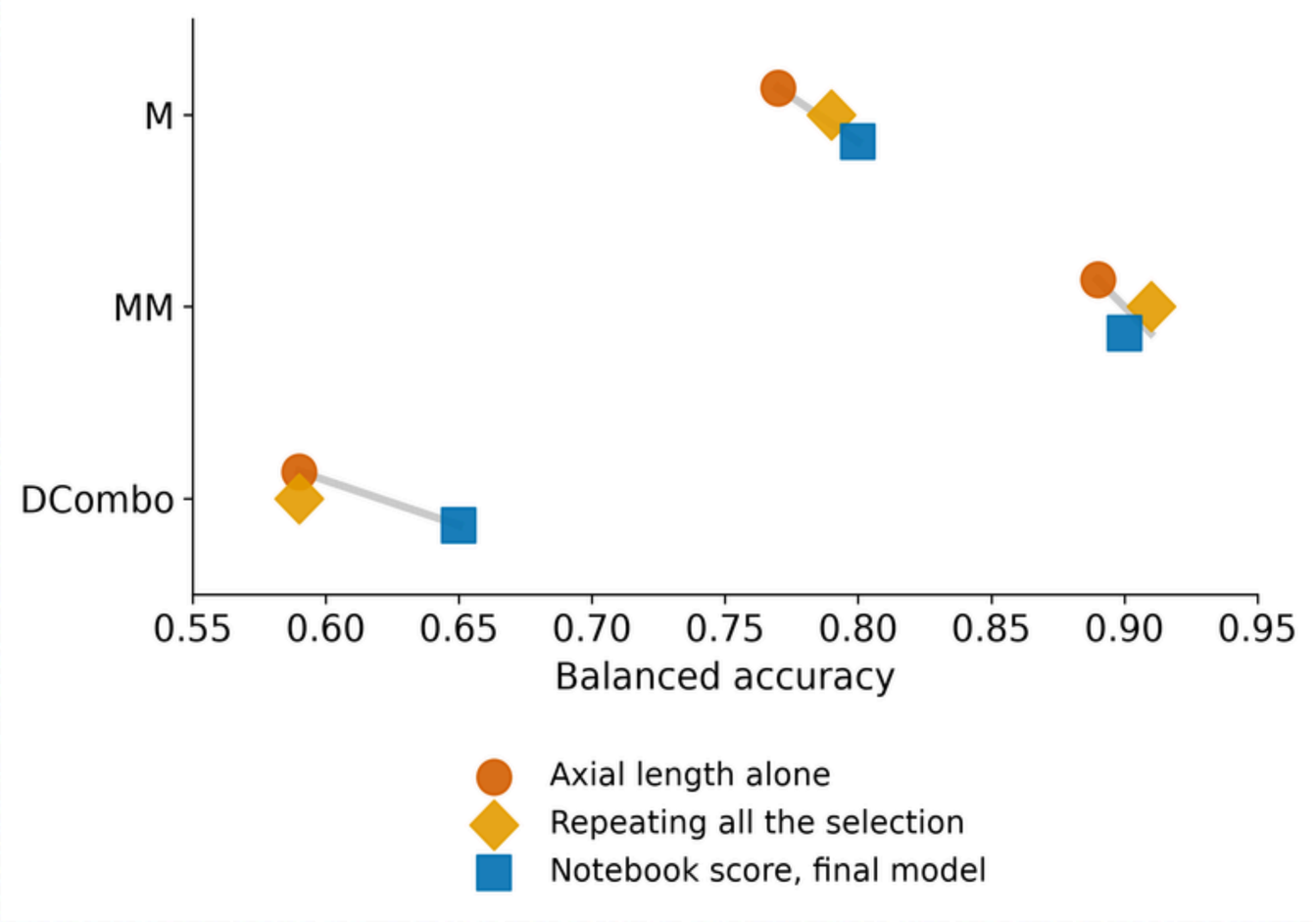
1 in 3 DCombo predictions off by one class

M1 the hardest class, often seen as control

1 in 6 magna cases missed by the MM model

Errors stay between neighbours

How optimistic are the scores



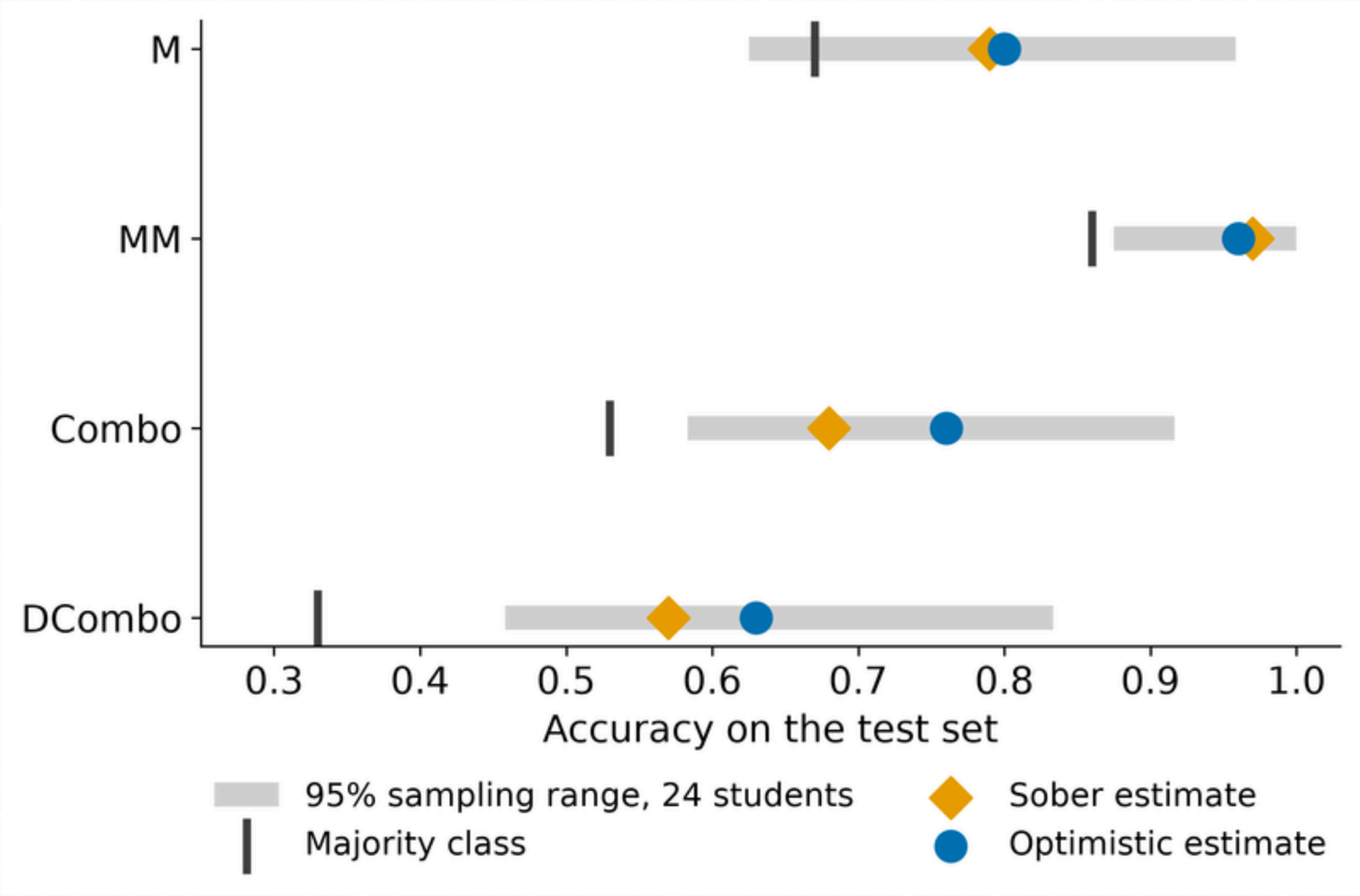
M the gain over axial length holds

MM the gain also holds

O gain left on DCombo and Combo

CV scores are an upper range

Expected test accuracy



24 test students: one is worth 0.04

4–5 students of spread from sampling alone

MM clearly above the majority class

One test score settles little

Conclusions

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01

Simple first

Axial length matched complex models on M and DCombo

02

Domain knowledge helps

The ratio feature delivered gains, most clearly on M

03

Measure the doubt

Nested checks show our scores are optimistic

Limitations

132 students
18 magna cases
No refraction values

Future work

More students
A direct refraction measure

TEAM CONTRIBUTION

Team and tools

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50/50

Tomás and Ana did all the parts together

Built with scikit-learn and XGBoost

CLOSING

Thank you for your attention

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THANK YOU